



A novel minimal distortion-based edge adaptive image steganography scheme using local complexity

(BEASS)

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Abstract

The advantage of spatial domain image steganography techniques is their capacity to embed high payloads of data by directly modifying image pixels. While these techniques have a high-embedding capacity, they often create visual and statistical distortion in smoother regions. Most existing edge steganography techniques divide an image into blocks and insert data by processing the blocks in a linear order, but these method also has multiple drawbacks. First, if the selected block has an insufficient number of edge pixels, it may result in multiple blocks being processed. Second, at high embedding rates, the method creates severe distortion as multiple message bits are hidden in edge pixels and surrounding non-edge pixels without analyzing the statistical dependencies and correlation of pixels, compromising data security. The aim of the proposed method is to construct a *Block-wise Edge Adaptive Steganography Scheme (BEASS)* using textured regions, particularly edges and surrounding pixels. This scheme dynamically chooses the region to embed messages using a local complexity measure of *Standard Deviation*. It offers high payload, minimal distortion embedding by hiding three message bits into edge pixels using the minimal *Mean Square Error* to determine the embedding capacity of neighboring non-edge pixels within the block to preserve the statistical dependencies. The practical merit of this approach was validated and compared with existing algorithms, and experimental results find that the proposed method surpasses IQM tests, achieves a high PSNR of $61\sim 65$, proves to be robust against kurtosis and skewness distortion, resists histogram attack, RS steganalysis and high dimensional ensemble classifier at *80% block modifications*.

Keywords Adaptive image steganography · Image Quality Metrics (IQM) · Least Significant Bit (LSB) · Local complexity analysis · Statistical distortion · Mean Square Error (MSE) · Steganalysis

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1 Introduction

Image steganography is the art and science of covert communication in digital images. Generally, the image in which we intend to embed secret data is called the *cover* image, and the resultant image after data is embedded is called a *stego* image. The design of any steganographic scheme should adhere to three principal requirements: first, ensure high embedding capacity; second, maintain high imperceptibility; and third, provide maximal robustness against steganalytic attacks attempting to break the scheme. Unfortunately with increase in payload, visual and statistical distortion in the form of noise will be introduced into the *stego* image, making them highly susceptible to attacks [1]. Due to this, past and current image steganographic systems typically tradeoff high-embedding capacity to maintain imperceptibility so that the resultant stego is as identical to the cover image as possible [4, 21, 27]. Adaptive image steganography approaches take advantage of reduced human sensitivity to changes in complex regions and carefully considers the inter-pixel variations to estimate a safe embedding payload [29]. Textured regions comprising of edges are better suited to adaptive steganography because it resists common *steganalysis* attacks – especially as the noise present in edges are difficult to model due to sudden, sharp change in intensities and statistical properties [12, 14]. Although edge based adaptive techniques ensure imperceptibility and high robustness, it is limited by the following issues:

1. **Number of edge pixels:** Most cover images consist of few edge pixels, hence their embedding capacity is relatively lower than other spatially-adaptive steganography techniques, resulting in a lower payload capacity [22].
2. **Imprecision of edge recognition:** Conventional edge detectors such as Sobel, Robert, Prewitt and Canny cannot recognize the edge regions precisely in the cover and stego images causing disparities in the extraction process [20].
3. **Statistical dissemblance:** In order to achieve high payload embedding, the majority of schemes [22][2, 11, 13, 28] modify three to five bits of the edge pixels to accommodate the payload - consequently reducing emphasis on the statistical resemblance of cover and stego images.

This paper focuses on refining block-wise edge adaptive steganography and aims to address the issues above by introducing a novel, minimal distortion, edge image steganography scheme called ***Block-wise Edge Adaptive Steganography Scheme (BEASS)***, that adaptively adjusts itself according to the message payload by choosing regions with the highest local complexity to embed data. Using this method, the cover image is divided into regions of 64×64 pixel blocks and an accurate estimate of the local complexity of the image is determined, wherein a high complexity value identifies textured regions with the maximum number of edge pixels that can be modified. To achieve high payload embedding, the selected region is further divided into sub-regions of 3×3 blocks and three message bits are hidden in the edge pixels, while simultaneously determining the embedding capacity of the surrounding non-edge pixels to preserve the image's statistical resemblance. The rest of the paper is organized as follows: Section 2 analyzes the mechanism and limitations of related approaches; Section 3 discusses the embedding and extraction algorithm of the proposed scheme in detail; followed by experimental results and comparisons in Section 4; and finally, in Section 5, we conclude our work.

2 Related work

The simplest and widely used Least Significant Bit (LSB) embedding methods are LSB Substitution (LSBS), also known as LSB Replacement (LSBR) [6] or LSB Matching (LSBM) [7, 17] which can hide messages in random LSB bits determined by a PRNG generator. LSBR works by replacing or substituting bits in the first bit-plane while preserving the higher (2~8) bit planes, making it possible to hide large message bits into the LSBs of the cover image. LSBM ensures at most ± 1 random bit change whenever the message bit is not equal to the LSB of cover pixel. It requires boundary values analysis [0, 255] of pixels after modification of for grayscale images. Jarno Mielikainen proposed a variation of LSBM called LSBM Revisited (LSBMR) [25] which uses a pair of adjacent cover pixels (i_i, i_{i+1}) pair to hide one bit in the first pixel $LSB(s_i)=(m_i)$ and another in a binary function $LSB(\frac{s_i}{2} + s_{i+1})=m_{i+1}$ of the two pixels to obtain the corresponding stego pixels (s_i, s_{i+1}) where m_i and m_{i+1} are the two message bits. LSBMR has the same embedding capacity as LSBM with fewer cover pixel changes. Statistical deterioration was not avoided as modifications were spread throughout the image, including smoother regions.

The motive behind using edges in traditional and current image steganography is lower distortion as image modifications go undetectable by the human visual system. The efficiency of edge-based steganography is contingent on the number of edges, where fewer edges result in loss of message information. Luo et al. [24] expanded LSBMR and proposed an adaptive edge embedding using LSBMR called AE-LSBMR; their focus was to preserve the statistical and visual correlation of pixel pairs, where two bits are carried according to four conditions. Gradients or the pixel pair difference, were used to distinguish the complexity of the edges based on a threshold which was adjusted according to the payload. Initially, the message was carried in the sharper edges and as embedding rate increased, more edges were utilized as carriers. Most of the adaptive scheme focuses on minimizing distortion and preserving the statistical characteristics of the cover but embeds relatively low payload.

The use of classical edge detectors like Sobel, Prewitt, Robert, Laplacian and Canny [3, 5] for identification of edges have been employed in multiple techniques, of which Canny edge detector is the most popular. It was found that these classical detectors were unable to identify the exact edges in the cover image, which in turn caused variations in the edges identified in the stego image, ultimately leading to incorrect information retrieval.

Another approach for edge identification is Semantic edge detection (SED), which aims to extract edges as well as its category information. Liu Yun [30] developed an SED approach using a fully convolutional neural network with diverse deep supervision (DDS). The lower layer of their multi-task framework approach generated category-agnostic edges, while higher layers detected category-aware semantic edges. SED has been extensively researched for application domains such as semantic segmentation, object proposal generation and object recognition but it has not been adopted to edge information embedding. Liao et al. [23]'s method was based on adaptive color image steganography aimed to exploit inter-RGB channel correlations to allocate payload for performance enhancement. The embedding capacity among the RGB channels was determined by amplifying channel modification probabilities using distortion costs for each pixel. This was done to obtain the save payload for each channel after which data is split and embedded into segments of the channel. It was evident that the costs of edge pixels would be considerably less than pixels at the smoother region which should ideally have higher payload capacity but since, the payload is computed according to channel, the edge pixels tolerance to modification were not completely utilized.

In [20], Sanjeev et al used a set of fuzzy conditions to identify the sharp and weak edges. The conditions were based on sharp intensity changes of edges in cover image compared to the homogenous continuous areas. At high payload, both the sharp and weak edges were used adaptively, whereas for smaller payloads changes were restricted to only the sharper edges. They tested the Fuzzy Inference System (FIS) for edge steganography and it was found to be most effective in identifying the exact edges in both the cover and stego image, thereby eliminating the discrepancy in message retrieval. Only two message bits were hidden in the LSB pair of each edge using LSBS embedding. The visual quality of the stego was observed to be as good as the cover image indicated with high PSNR value. However, the embedding was carried at only two edge bits so the payload hiding capacity was low. Additionally, if the chosen cover image was not as textured, the algorithm would choose the weaker edges to fulfill the payload requirement.

Edge techniques targeted to meet high payload embedding such as in [28], Chen et al. proposed a block-based method with a hybrid edge detector to determine the edge pixels. The first block P1 acts as the indicator which carries information about the edge and non-edge pixels in the other blocks P2, P3, and P4. Then, two message bits in the non-edge pixels and four message bits in the edge pixels of P2, P3, and P4 blocks were embedded. Chen et al. made a substantial improvement in terms of embedding capacity but the scheme has not considered the dependencies of the pixel within the block that cause visible artifacts or distortions as indicated by a low PSNR value.

Tseng et al. [13] revisited Chen's method to minimize the distortion caused due to embedding changes but their method also fails to utilize the available space completely as a large amount of the space was wasted in storing non-edge information; it also has high computational time. Their method divided the cover into 4×4 blocks which was further divided into 2×2 blocks namely UL, UR, LL, LR,. The first block, UL with subblocks P1, P2, P3, P4 acts as the indicator to store the embedding length of the edge and neighboring non-edge pixels which could be any of the four cases $[x, x+1]$, $[x, x+2]$, $[x, x+3]$, $[x, x+4]$. The number of edges and non-edge information of the UR, LL and LR blocks is stored in four LSBs of P2, P3 and P4 respectively. For example, if UR block had two edge and two non-edge pixels then the last four LSB of P2 was '1010' scanned in zig-zag manner. In these cases, the first parameter x was the number of bits to hide in non-edge pixels and the second parameter was the bits to hide in the edge pixels; among these cases, the one with the *Minimum Square Error (MSE)* is chosen as the embedding unit for the block.

More recently in [2], Junlan et al. proposed a scheme to overcome the computational complexity and the wastage of embedding space by indicator blocks as observed in Tseng's scheme. Their focus was to reduce the embedding distortions by ensuring that the stego images had high PSNR values. Their method reserved five bits for embedding, it began by clearing five LSBs of the original cover image to generate a secondary image which was fed as an input to Canny, Sobel and Fuzzy edge detection system. They implied that the edge image generated using the remaining three Most Significant Bit (MSB) in the secondary image was as good as the edge image generated using the original image. During the embedding process, the edge image was divided into 2×2 blocks and two bits were embedded in non-edge pixels and four bits in edge pixels to produce the stego image using LSBS. Their algorithm claims to produce more edge pixels using the edge detectors than any other approach discussed previously. During extraction, another secondary stego image was generated to produce the stego edge image, from which message bits were extracted from the edge and neighboring non-edge pixels. Junlan et al. does not inspect the statistical distributions of the cover/stego images and the correlations between pixels within the block - resulting in easy detection. Moreover, there was no substantial evidence of correct edge

detection using the secondary image at the receiver's end, which may cause discrepancies limiting the scope of the method.

This study aims to overcome the limitations observed in current adaptive edge and block-wise edge approaches; it is also the motivation behind constructing the proposed minimal distortion, edge adaptive image steganography scheme - a simple yet effective method in the spatial domain for grayscale images. In particular, the proposed method intends to improve existing adaptive edge scheme's low embedding efficiency by embedding a safe payload in edge and non-edge pixels within a block, as determined by MSE, while preserving the local textures and statistical characteristics of the images throughout the embedding process. To reduce the block computational complexity, standard deviation is used to compute the local complexity; high complexity identifies blocks with largest number of sharp edged pixels as they can tolerate most of the modifications. In case of high embedding rates, the scheme is designed to gradually add blocks with weaker edges.

3 The proposed method

The paper extends the block-wise information hiding approach by using local complexity features to determine the inter-pixel dependency and to identify complex regions. In Fig. 1a, a flow diagram illustrates the data embedding process; the edge image is constructed using an FIS detector that locates the edges accurately and the block complexity of the selected region is determined. The next step is to estimate the embedding capacity of edge and neighboring pixels and finally perform the data embedding process. The algorithm automatically selects the next region and repeats the embedding process as needed. The proposed method requires the support of three secret-keys (integers) to guarantee the validity of data extraction. It requires 128 bits to represent the integers if we adopt the 32 bits IEEE 754 single precision representation. Alternatively, approximately four sub-blocks can be forfeited to represent side-information. The side- information can be embedded into the cover image.

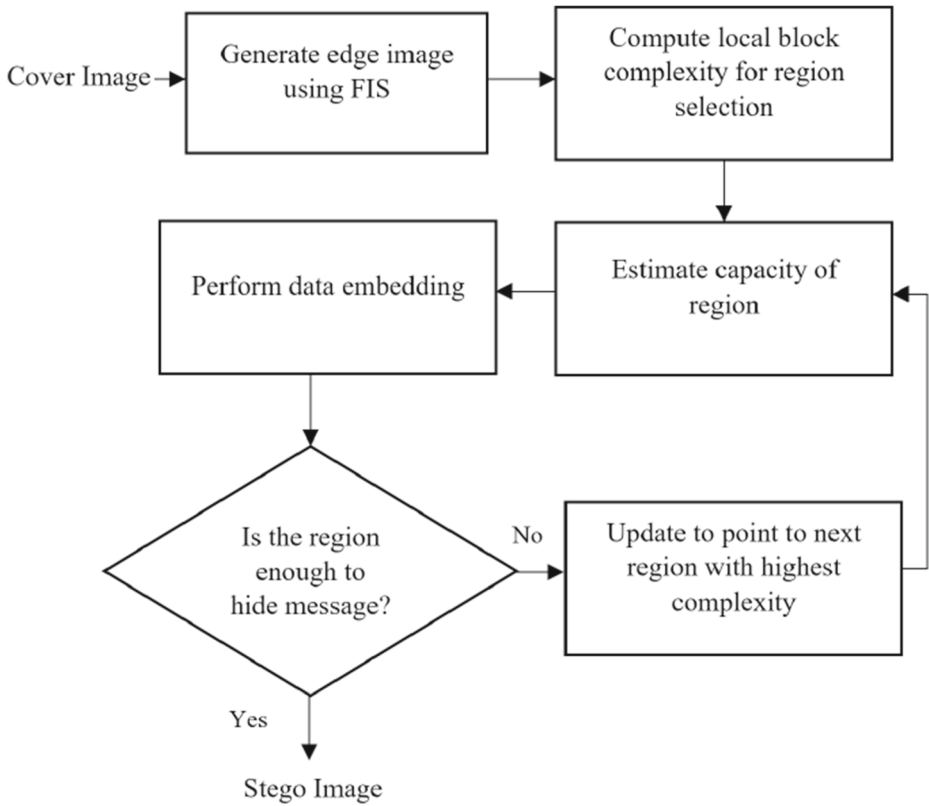
1. Secret Key-I: To determine the order of blocks used for data embedding.
2. Secret Key-II: To notify the block size used.
3. Secret Key-III: To notify the sub-block size used.

In the data extraction phase illustrated in Fig. 1b, the keys or the side information is retrieved, following which the receiver performs a similar operation in the data embedding phase to generate the stego edge using FIS before being divided into blocks; then depending on the block order received, it selects the appropriate region and applies the corresponding extraction algorithm to retrieve the message. Subsequent sections provide a detailed explanation of the phases.

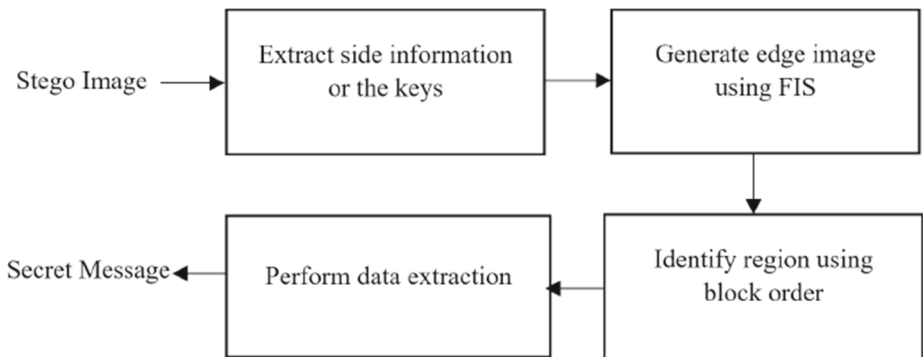
3.1 Data embedding procedure

3.1.1 Fuzzy edge detectors

The construction of the edge image is a crucial step in all edge image steganography. The edge detector must be able to determine the exact edge regions in the cover image. During the embedding process, the edge pixels detected should be large enough to carry high degree of modification. We have used Fuzzy Edge detector (<https://in.mathworks.com/help/fuzzy/fuzzy-logic-image-processing.html>) and briefly reiterate the FIS edge detection method below:



(a)



(b)

Fig. 1 Proposed BEASS Method **a** Data Embedding Procedure **b** Data Extraction Procedure

1. Breaks in uniform region is identified by image gradient, I_x along x-axis and I_y along y-axis; I_x and I_y is obtained by convolving image with gradient filters $[-1, 1]$, and $[1, -1]^T$ respectively.
2. I_x and I_y with zero-membership functions are fed as inputs to the Fuzzy Inference System (FIS) for edge detection; the performance of which is dependent on the Standard Deviation values.
3. The intensity of the edges selected is defined using a Triangular membership function such that white is denoted by $[0.1 \ 1 \ 1]$ and black by $[0 \ 0 \ 0.7]$.
4. Two very simple rules stating
 - if (I_x and I_y)=0, then output=white
 - if (I_x and I_y)!=0, then output=black
5. Finally, evaluate the FIS using I_x and I_y row wise to obtain the edge image.

In Fig. 2, the Fuzzy FIS edge image generation using sample cover images from BOSSbase 1.0 database at 0.6 threshold is illustrated. Clearly, the choice of cover image should be textured enough to completely accommodate the secret message.

3.1.2 Estimate local block complexity for region selection

Most block-wise edge steganography schemes process blocks in a linear manner. Here, blocks refer to regions containing edges. Circumstantially, if the embedding modification

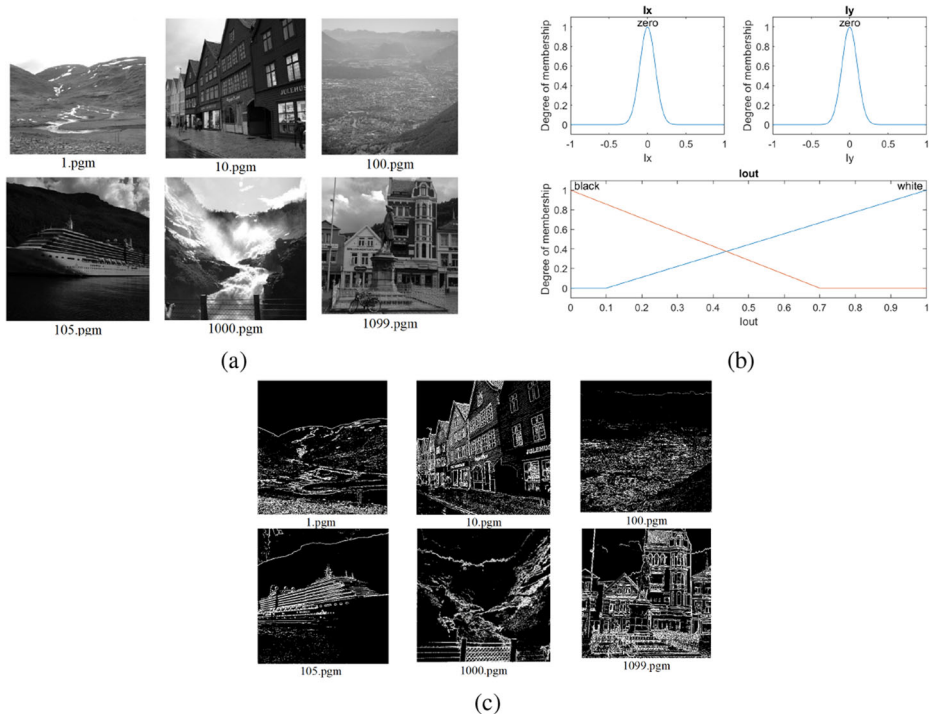


Fig. 2 **a** Some sample cover images from Bossbase 1.01 database used throughout the proposed BEASS scheme **b** The FIS membership function I_x , I_y to identify breaks in the cover image and Triangular function to specify the intensity of the identified edges **c** The generated six edge images

starts at blocks with insufficient or weak edges then multiple blocks are processed to accommodate the message, inevitably increasing computational time. On the contrary, if embedding modifications begin at blocks with the maximum number of sharp edges, then the block processing overhead will be reduced as the message can be accommodated with fewer modifications. Also, the texture characteristics of the original cover image will be preserved as alterations at edges are often ignored by the HVS. Using the proposed method, the block-based approach is expanded to address the issue of region selection by statistically analyzing the cover to select blocks with the highest local block complexity. The region algorithm is incremental and adaptive in the sense that it selects the next block with highest value until the entire message is completely embedded; it continues selecting regions with the maximum sharp edges before selecting weaker edge regions in case of high payloads. The steps for region selection are as follows:

1. The cover image, I of $M \times N$ pixels is divided into 'b' number of non-overlapping blocks of $n \times n$ pixels. For our study, grayscale images of 512×512 pixels are used with $b=64$ blocks of 64×64 pixels.
2. The local block complexity is determined by the standard deviation (SD) of the each block, $b_i=1,2,...,m$ using (1) where m_i is the mean of pixel values within block b_i .

$$\sqrt{\sum_{i=1}^{n \times n} \frac{(X_{ij} - m_i)^2}{n \times n - 1}} \quad (1)$$

where X_{ij} denotes each pixel within block b_i .

3. The order of embedding regions is obtained by sorting the block standard deviation values in decreasing order such that $SD_i \geq SD_{i+1} \geq SD_{i+2}$. High standard deviation represents the region that is highly textured and contains the most edges. The algorithm picks the first block and the next block from the order if the payload demands it. Taking image 10.pgm as an example in Fig. 3, the algorithm selects five regions out of the 64 regions (blocks) containing maximum strong edges when 10% block modifications are to be made.

3.1.3 Compute capacity estimation and embedding method

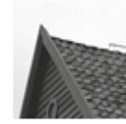
This section elucidates the data embedding procedure performed on one region at a time using the *Least Significant Bit Substitution (LSBS)* method. During the actual data embedding, the selected region (block) of 64×64 pixels is divided into sub regions (subblocks) of 3×3 pixels to estimate the embedding capacity using minimum *Mean Square Error (MSE)*. Only the sub region containing edges are considered for calculation; smooth regions with no edge pixels are exempted. The sub region MSE is calculated using (2).

$$MSE = 9 \sum_{i=1}^3 \sum_{j=1}^3 (I'(i, j) - I(i, j))^2 \quad (2)$$

where $I(i, j)$ is the cover pixel and $I'(i, j)$ is the modified pixel value at (i, j) location. Within the subblock, the proposed method uses an embedding length of edge pixels as three and of its neighboring non-edge pixels as one or two, depending on their estimated capacity. This is because edges can tolerate more modifications than non-edge pixels, only causing an acceptable amount of visual and statistical distortion. It should be emphasized capacity estimation for the block is crucial since embedding a lesser amount of data does not always



(a) Region 1



(b) Region 2



(c) Region 3



(d) Region 4



(e) Region 5

Fig. 3 Region selection for image 10.pgm based on local block complexity. Applying 10% embedding change will result in selection of five blocks with highest standard deviation values as regions

guarantee the minimum MSE and the overall payload hiding capacity would also be undesirably less [13]. As shown in Fig. 4, to clearly explain the process of message hiding, the following variables are used to represent pixel bits and the embedding procedure within each subblock (3×3 pixels).

1. Let the 8-bit pixel positions be denoted by ' $b_7b_6b_5b_4b_3b_2b_1b_0$ '.
2. Let sub block pixels be denoted by $P_1, P_2, \dots, P_9$
 - where the top left pixel ' P_1 ' stores the embedding length of non-edge pixels
 - For example, if the non-edge pixel embedding length is one, then ' b_0 ' of ' P_1 ' is set to 0
 - If embedding length is two, then ' b_0 ' of ' P_1 ' is set to 1. Then, secret message bits are embedded into $P_2, \dots, P_9$ according to the pixel type.
3. Let ' s ' denote the non-edge embedding capacity of the subblock.
4. Let ' $bitset$ ' be 0 or 1, to be stored in ' P_1 ' to indicate ' s ' value.
5. Let ' $MSE1$ ' denote the MSE value when edge pixels hide three message bits in ' $b_2b_1b_0$ ' of edge pixel and one message bit in ' b_0 ' of non-edge pixel.
6. Let ' $MSE2$ ' denote the MSE value when edge pixels hide three message bits in ' $b_2b_1b_0$ ' of edge pixel and two message bits in ' b_1b_0 ' of non-edge pixel for the same message

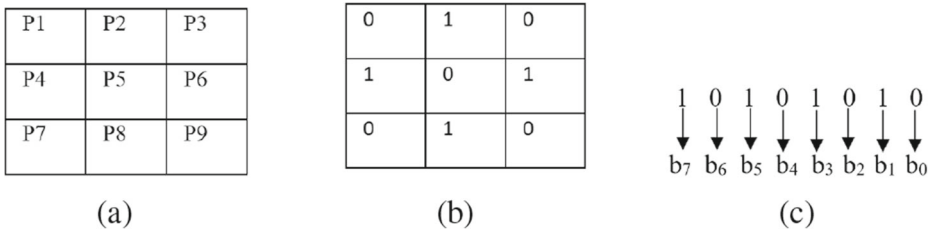


Fig. 4 **a** Subblock numbered P1 to P9 **b** Edge subblock, 1 indicates edge pixel and 0 indicates non-edge pixel **c** 8-bit pixel notation used in the proposed scheme

bit stream. The length is chosen such that statistical distortion is minimal and overall payload capacity is not compromised.

The embedding procedure is summarized below:

1. Generate edge image given the cover using FIS as shown in section (see Sect. 3.1.1).
2. Estimate local pixel complexity for region selection as explained in section (see Sect. 3.1.2).
3. Fetch the selected region (block) of 64×64 pixel for data embedding and divide into 3×3 pixel sub regions (subblocks).
4. Select subblock with edge pixel, determine the embedding capacity, 's' of the non-edge pixel. The algorithm chooses to embed 1 bit only when we see significant difference ≥ 1 .
 - if $MSE2 \leq MSE1$ then,
 - $s=2$
 - else $(MSE2-MSE1) \geq 1$ then,
 - $s=1$
5. Set the top-left pixel, P1's last bit b_0 to 0 if $s=1$ otherwise set to 1 to indicate $s=2$. Then for every edge pixel hide three message bits into the last three bits ' $b_2b_1b_0$ ' and hide 's' message bits into the last s bit ' $b_s, \dots, b_0$ ' of the non-edge pixel using LSBS embedding.
6. If there are message bits still to be embedded, fetch the next subblock and repeat step 4 and 5.
7. After processing the block, if there are message bits remaining, then fetch the block with the next highest standard deviation value and repeat the procedure from step 2 to 6 until the entire message is successfully hidden in the cover image.
8. Finally, the stego image is generated.

The working of the algorithm, after dividing the block with highest standard deviation into sub blocks, is demonstrated in Fig. 5. Consider the two cases; Case I when $(MSE2-MSE1) \geq 1$ and Case II when $MSE2 \leq MSE1$. First, 'MSE1' and 'MSE2' is evaluated. To do so, cover sub block's 'P1' pixel's b_0 is substituted with '0' as shown in 'Sub block MSE1' and 1 in 'Sub block MSE2'. Then ' $b_2b_1b_0$ ' bits of 'P2, P4, P6' (edge pixels) are substituted with three message bits using LSBS and b_0 bit of 'P3, P5, P7, P8, P9' (non-edge pixel) are substituted with one message bit using LSBS - to compute 'MSE1'. Likewise, to compute 'MSE2' b_1b_0 bits of 'P3, P5, P7, P8, P9' (non-edge pixel) are substituted with two message bits using LSBS.

Once the MSE values are known, determine the case, check ' $(MSE2-MSE1) \geq 1$ ', if true, then it falls under 'Case I' and 'bitset' value is set to '0' to denote ' $s=1$ '; If false, then it falls under 'Case II' and 'bitset' value is set to '0' to denote ' $s=2$ '. Finally, the stego

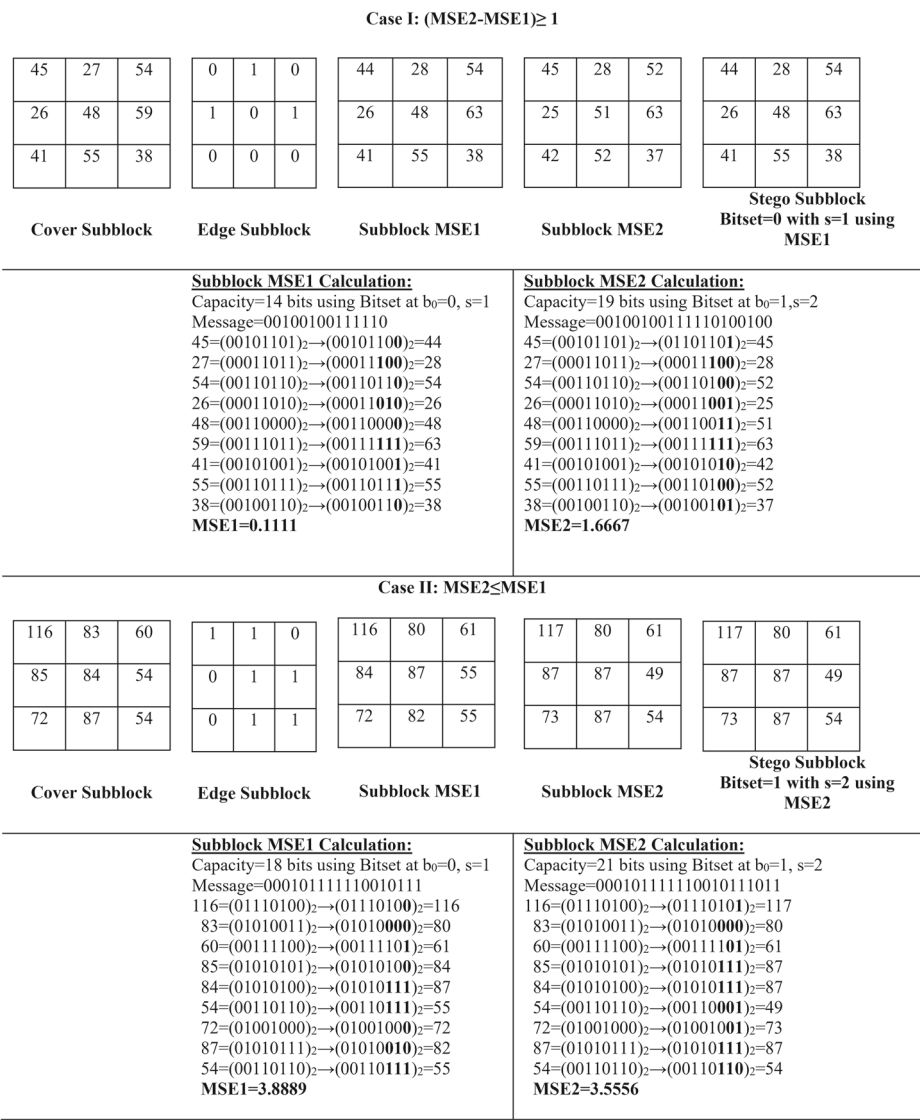


Fig. 5 Case I and Case II of the proposed embedding procedure

subblock is obtained by embedding three message bits and ‘s’ bits into the non-edge pixels. The side information, as stated previously, consists of the three secret keys. Blocks in cover image are not used to communicate the side information to the receiver, instead the secret keys are conveyed through secured channel. Once this information is received, the extraction procedure begins at the receiver’s end. The stego images generated by BEASS using cover image 10.pgm, 1000.pgm and 1099.pgm when exposed to 10% - 95% block modifications (0.3 - 1.6 bpp) are indistinguishable from the cover images as shown in Fig. 6, indicating excellent perceptual quality. The proposed BEASS method is advantageous as it can use up to 61 blocks at ~95% block modifications and still achieve excellent visual similarity at tolerable statistical distortion.

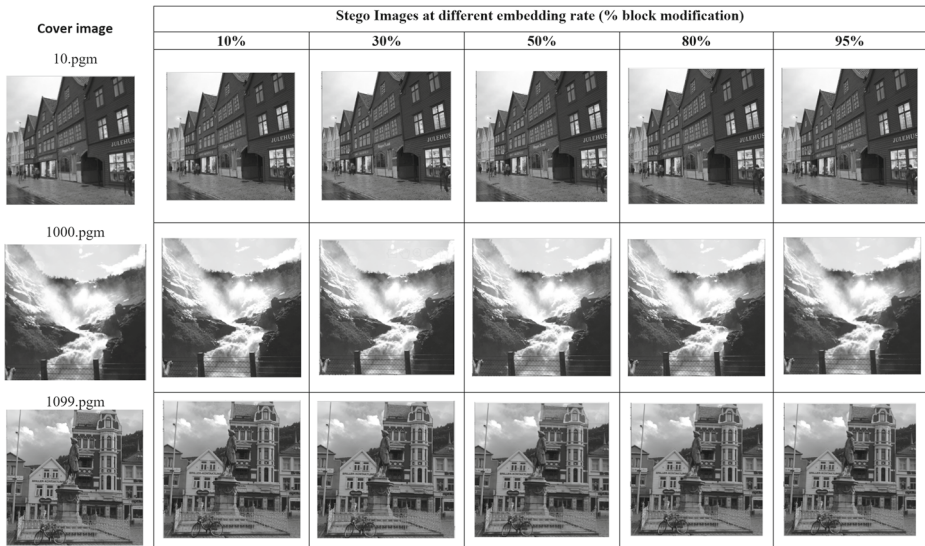


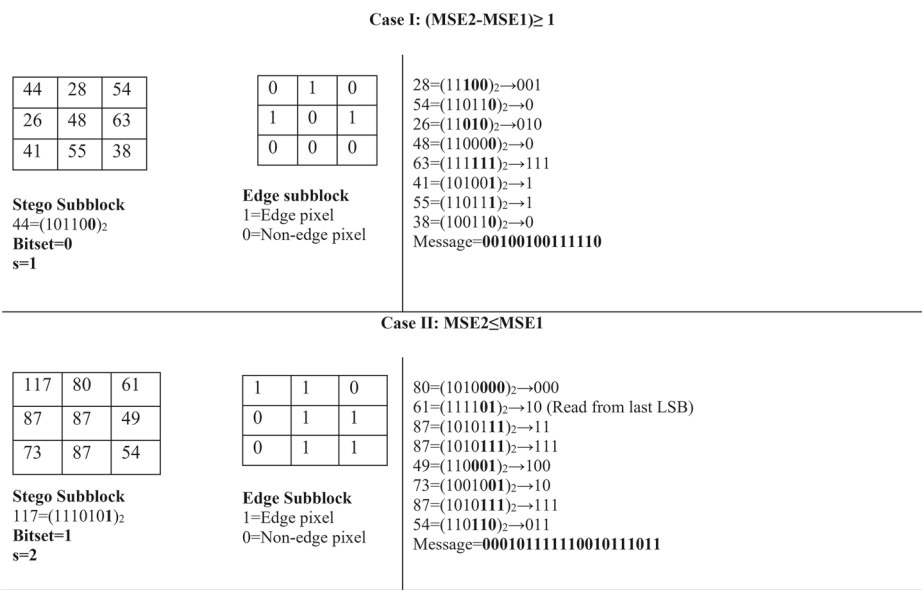
Fig. 6 Stego images generated using the proposed method exposed to 10% to 95% block modifications

3.1.4 Data extraction procedure

The validity of the message extracted from the stego image is dependent on the three secret keys shared by the sender. They are denoted with variables for easy reference; Secret Key-I is order of blocks (o) used in data embedding, Secret Key-II is block size (n), and Secret Key-III is sub-block (k). Using the keys, the extraction algorithm can now extract the message bits as specified below:

1. Generate edge image of the stego using FIS as shown in section (see Sect. 3.1.1).
2. Divide the stego image into 'b' blocks of $n \times n$ pixels.
3. Until message 'm' is extracted from the blocks:
 - Take the first block from the order 'o' and divide into subblocks of $k \times k$ pixels.
 - Fetch the sub block one at a time, determine the 's' value from the top-left pixel P1's last bit, b_0 .
 - Then, for every edge pixel, extract three message bits from $b_2b_1b_0$ and $b_s, \dots, b_0$ message bits from the non-edge pixels.
 - Fetch the next block from the order and repeat.
 - End
4. Finally, the message 'm' is extracted.

This is illustrated in Fig. 7. First, stego edge image is generated using FIS. Then stego and edge image is divided into non-overlapping blocks of 64×64 pixels ($n=64$). Using the first block from block order 'o', divide into subblocks of 3×3 pixels ($k=3$) - for both cases. Considering the 'stego sub block of Case I', obtain P1's b_0 to find the 'bitset' (which is 0 indicating 's=1'). To acquire the message, extract three bits $b_0b_1b_2$ from edge pixel and b_0 from non-edge pixel. In case II, P1's b_0 is extracted to find the 'bitset' (which is 1 indicating 's=2'). Then, extract three bits $b_0b_1b_2$ from edge pixel and b_0b_1 from non-edge



technique which embeds on both sharp and weak edges according to payload; *Adaptive Fuzzy Edge* offers high payload embedding by modifying two pixels in edge bits as detected by a fuzzy system; *high payload block edge* is the most recently proposed high payload, block based, edge embedding method. The images used throughout the test is taken from BOSSbase 1.01 database (<http://dde.download/ImageDatabase.binghamton.edu>) which is a well-known, benchmarked dataset of images obtained from a digital still camera consisting of 10,000 RAW, 512×512 grayscale images in .pgm format.

4.1 Image quality analysis

The visual and statistical image quality metrics (IQM) [21][27] (<https://in.mathworks.com/matlabcentral/fileexchange/image-quality-measures>) used to evaluate and compare are the *Peak Noise to Signal ratio (PSNR)*, *Structural Information Change (SSIM)*, *KL-Divergence*, *Average Difference (AD)*, *Normalized Absolute Error (NAE)* and *Execution Time*.

The **Peak Noise to Signal ratio (PSNR)** calculated using (4) is a subjective quality based on HVS characteristics to differentiate cover and stego images. The MSE value calculated using (5) used in PSNR will be evaluated on the overall image unlike MSE for subblock.

$$PSNR = 10 \times \log_{10} \left(\frac{255 \times 255}{MSE} \right) \quad (4)$$

$$MSE = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N (S(i, j) - I(i, j))^2 \quad (5)$$

where cover and stego image is represented by $I(i,j)$ and $S(i,j)$ respectively and size of both the image is $M \times N$ for $i=1,...,255, j=1,...,255$.

As an example, Table 1 shows in comparison with other methods, the proposed BEASS method achieves the highest PSNR value in the range of [70.00 - 61.00 db] for different embedding rates (0.3, 0.5, 0.8 bpp), calculated for image 10.pgm, 1000.pgm and 1099.pgm. This indicates BEASS method passes the imperceptibility test as the stego images cannot be detected by the HVS. The resultant PSNR values are obtained using maximum of 51 blocks to equally compare with the other methods.

The **Structural Information Change (SSIM)** calculated using (6) is the similarity measure between cover and stego image based on combination of three factors that are loss of correlation, luminance distortion and contrast distortion. It offers a good estimation of perceived image distortion; the proposed BEASS method's stego image have no visual distortion and is structurally identical to the cover image as indicated by the SSIM value ($0.99 \leq x \leq 1$) calculated for image 10.pgm, 1000.pgm and 1099.pgm at varying embedding rates as seen in Table 1.

$$SSIM = \frac{2\mu_i\mu_j + C_1}{\mu_i^2\mu_j^2 + C_1} \cdot \frac{2\sigma_i\sigma_j + C_2}{2\sigma_i^2\sigma_j^2 + C_2} \cdot \frac{\sigma_{ij} + C_3}{\sigma_{ij} + C_3} \quad (6)$$

where μ_i and μ_j are mean values, μ_i^2 and μ_j^2 are variances, σ_{ij} is covariance of i,j . The constants $C_1=(K_1L)^2, C_2=(K_2L)^2$ taking $L=255, K_1=0.01$ and $K_2=0.03$.

KL-Divergence ($D_{KL}(P_c||P_s)$) is a security and similarity measure based on probability distribution cover and stego image. The BEASS method is able to achieve minimal KL-Divergence (ranging from 0.76e-05 to 1.37e-05) in comparison with other methods,

Table 1 Comparison of IQM parameters of the proposed BEASS algorithm with existing adaptive edge, high payload algorithms

Cover Image	Algorithm	Embedding rate	PSNR	KL Divergence	SSIM	AD	NAE	Execution time
10.pgm	AE-LSBMR [24]	0.3	64.91	1.29e-05	0.999	0.1005	0.0010	0.71
		0.5	57.14	1.93e-05	0.998	0.1503	0.0014	1.21
		0.8	53.73	1.97e-05	0.996	0.4612	0.0029	3.61
		1.0	51.87	1.99e-05	0.995	0.9082	0.0041	5.75
	Fuzzy Edge [20]	0.3	61.12	1.64e-05	0.997	0.0017	0.0059	1.93
		0.5	53.68	1.69e-05	0.994	0.0852	0.0099	4.47
		0.8	52.29	1.78e-05	0.988	0.8003	0.0152	7.00
		1.0	51.28	2.36e-05	0.982	1.0002	0.0202	10.41
	Block Edge [2]	0.3	57.29	1.47e-05	0.996	0.0015	0.0013	2.08
		0.5	50.14	1.79e-05	0.995	0.0004	0.0020	7.69
		0.8	47.54	1.81e-05	0.993	0.0006	0.0035	8.40
		1.0	44.80	1.83e-05	0.989	0.0031	0.0059	9.32
	BEASS method	0.3	70.54	1.13e-05	0.999	0.0017	0.0002	1.39
		0.5	65.04	1.25e-05	0.999	0.0213	0.0008	1.98
		0.8	63.75	1.27e-05	0.998	0.0586	0.0017	2.68
		1.0	62.91	1.33e-05	0.997	0.0835	0.0031	4.83
1000.pgm	AE-LSBMR [24]	0.3	65.42	0.99e-05	0.999	0.0004	0.0001	0.59
		0.5	57.43	1.39 e-05	0.997	0.0029	0.0007	1.82
		0.8	55.36	1.66 e-05	0.993	0.0037	0.0012	3.35
		1.0	53.81	1.73 e-05	0.989	0.0090	0.0017	5.18
	Fuzzy Edge [20]	0.3	61.12	1.80e-05	0.993	0.0009	0.0001	2.02
		0.5	56.24	2.00e-05	0.991	0.0028	0.0008	5.55
		0.8	55.19	2.15e-05	0.987	0.0034	0.0019	7.89
		1.0	54.18	2.37e-05	0.982	0.0122	0.0033	9.97
	Block Edge [2]	0.3	61.33	1.02e-05	0.994	0.0016	0.0001	2.17
		0.5	53.11	1.28e-05	0.991	0.0070	0.0010	6.31
		0.8	50.63	1.34e-05	0.988	0.0140	0.0015	8.43
		1.0	48.43	1.56e-05	0.982	0.0212	0.0022	10.04
	BEASS method	0.3	74.66	0.76e-05	1	0.0004	0.0001	1.48
		0.5	68.82	0.89e-05	0.999	0.0012	0.0005	1.84
		0.8	66.48	1.11e-05	0.998	0.0025	0.0010	3.68
		1.0	65.78	1.23e-05	0.996	0.0096	0.0012	5.82

indicating good visual quality and security against attacks as indicated in Table 1. It is calculated using (7)

$$D_{KL}(P_c||P_s) = \sum_{i=1} (P_c(i) \log \frac{P_c(i)}{P_s(i)})$$

(7)

where P_c and P_s are probability distribution function (PDF) of the respective cover and stego images. $D_{KL}(P_c||P_s)$ is the information lost when P_c is used to approximate P_s .

Table 1 (continued)

Cover Image	Algorithm	Embedding rate	PSNR	KL Divergence	SSIM	AD	NAE	Execution time
1099.pgm	AE-LSBMR [24]	0.3	66.68	1.27e-05	0.999	0.0009	0.0010	0.76
		0.5	56.92	1.31e-05	0.997	0.0011	0.0013	0.93
		0.8	54.10	1.56e-05	0.995	0.0037	0.0027	3.49
		1.0	52.33	1.71e-05	0.992	0.0065	0.0039	5.80
	Fuzzy Edge [20]	0.3	63.35	2.06e-05	0.998	0.0014	0.0011	2.58
		0.5	53.36	2.10e-05	0.996	0.0030	0.0015	6.82
		0.8	52.30	2.17e-05	0.993	0.0089	0.0032	9.78
		1.0	51.13	2.21e-05	0.989	0.0161	0.0026	11.67
	Block Edge [2]	0.3	50.47	1.22e-05	0.996	0.0032	0.0013	5.42
		0.5	48.61	1.25e-05	0.995	0.0035	0.0051	10.57
		0.8	45.74	1.37e-05	0.993	0.0056	0.0067	13.56
		1.0	43.29	1.57e-05	0.990	0.0145	0.0089	15.55
	BEASS method	0.3	73.70	0.95e-05	1	0.0003	0.0009	1.87
		0.5	64.46	1.26e-05	0.999	0.0005	0.0011	2.60
		0.8	63.90	1.36e-05	0.997	0.0019	0.0024	3.53
		1.0	61.28	1.37e-05	0.996	0.0023	0.0029	5.78

The improved values against the respective parameters which shows the optimality of the method are shown in bold

Average Difference (AD) is an image quality to measure the change between the reference and the processed image and should ideally be zero to indicate maximum resemblance. The proposed BEASS method AD value lies in the range of 0.0003 to 0.0800 for the three test images as observed from Table 1. It indicates minimal difference between the cover and stego images signifying high similarity. It is calculated using (8)

$$AD = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N S(i, j) - I(i, j) \quad (8)$$

Normalized Absolute Error (NAE) is the measure of image visual quality, where in higher values indicates poor visual quality. From table 1, we observe negligible NAE values for BEASS method which cannot be detected with naked eye. It is calculated using (9).

$$NAE = \frac{\sum_{i=1}^M \sum_{j=1}^N |S(i, j) - I(i, j)|}{\sum_{i=1}^M \sum_{j=1}^N |S(i, j)|} \quad (9)$$

The execution time is another evaluation criteria to judge the proposed BEASS method as it is designed to reduce the block processing time. As a result of using local complexity to order the textured edge regions (blocks) the execution time is reduced by a significant factor in comparison with fuzzy edge [20] and block edge [2] but almost similar to AE-LSBMR [24]. This is because AE-LSBMR also uses an adaptive approach to identify edge regions according to payload.

Table 1 shows the summarized result of the IQM tests conducted on three test cover images ‘10.pgm’, ‘1000.pgm’ and ‘1099.pgm’. The tests starts with minimal block modification of 10% to highest of 80% (6 to 51 blocks), which is equivalent 0.3 to 1 bpp for all the compared methods. With further increase in payload, the proposed method can modify

up to 61 blocks at an equivalent to 95% change, exempting the last 3 blocks to impede the introduction of visible distortions. This was omitted in the test because for some of the comparative method, it caused severe visual distortion in stego image. Fig. 8 shows the graphical representation of BEASS with other methods on the IQM measures of PSNR, KL-Divergence, SSIM, Average Difference (AD), Normalized Absolute Error (NAE) and Execution time for image 1099.pgm obtained after performing 80% block modifications i.e. embedding rate of 1 bpp.

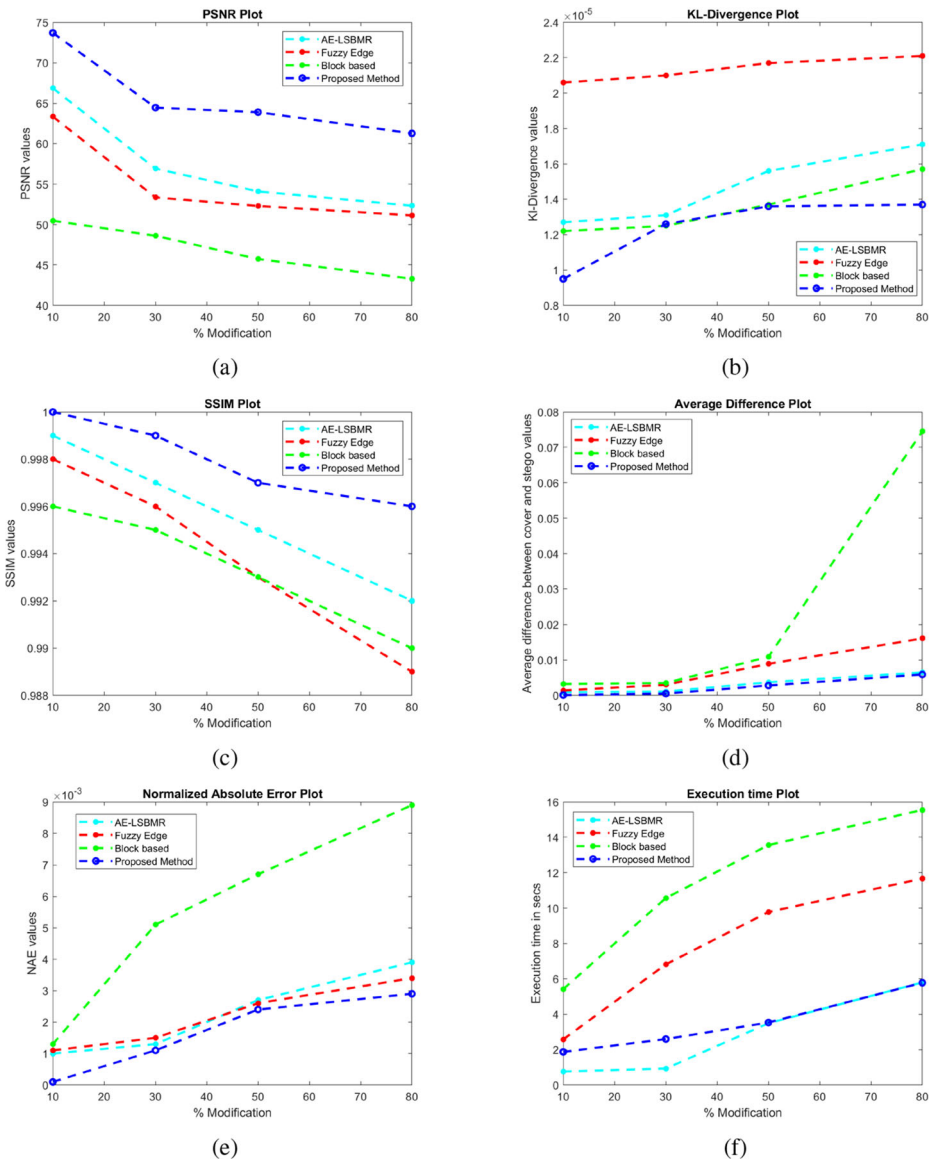


Fig. 8 Using image 1099.pgm, comparative analysis of IQM result at 80% block modification for BEASS i.e. 1 bpp on average for all the comparative methods

4.2 Security analysis

Security is an important parameter to consider in any steganographic systems [7]. The proposed method has been designed to reduce the statistical distortions introduced by the embedding process. To test, 13 blocks with least SD values are exempted and 51 blocks out of 64 blocks are used as image regions, which measures close to 80% of the available embedding blocks in the image. In such a case, when all the 51 regions are used for embedding, the embedding rate approximates to 1 bpp on average for all the comparative methods (AE-LSBMR [24], Sanjeev's Adaptive Fuzzy Edge [20] and Junlan's high payload block edge [2]) used in this paper. The evaluation is done using Kurtosis [9], Skewness [19], Histogram analysis [16], RS attacks [10], which are efficient steganalyzers for detecting spatial domain LSB steganography and feature based universal steganalyzer [15, 18].

4.2.1 Kurtosis and skewness

The shape of the image probability distribution can be further described by higher order image statistics called Kurtosis [9] and Skewness [19]. Kurtosis describes the 'peakedness' of the probability distribution of a real-valued random variable. Skewness, on the other hand, is an assessment of the 'asymmetry' of the probability distribution of a real-valued random variable about its mean value. For a univariate data $Y_1, Y_2, \dots, Y_N$

$$Kurtosis = \frac{\sum_{i=1}^N (Y_i - \mu)^4 / N}{\mu^4} \quad (10)$$

$$Skewness = \frac{\sum_{i=1}^N (Y_i - \mu)^3 / N}{\mu^3} \quad (11)$$

Where ' μ ' is the *mean*, ' σ ' is the *standard deviation* and ' N ' is the number of data pixels. The results in Table 2 scrutinizes the subsisting discrepancy between the residing components of the respective carrier and stego images against the third and fourth statistical moment of the distribution (skewness and kurtosis respectively) about the mean when applying embedding modifications of 80%. The estimated deviation is unnoticeable and thus the efficiency of the proposed approach is established. The proposed method shows that the kurtosis and skewness remain unchanged while for the other approaches, it varies. Thus, it can be rightfully said that BEASS method is secured against attacks which uses the distribution discrepancies in cover and stego images to detect hidden message.

4.2.2 Histogram analysis

LSB methods are highly susceptible to histogram attacks as it affects the last bit plane which distorts the histogram of the generated stego image [16]. In Fig. 9, a histogram analysis of BEASS is performed and compared after applying embedding modification in 80% blocks (approximately 1 bpp) on cover image 1099.pgm. The histogram of the cover image (highlighted in blue) and stego image (highlighted in red) have a negligible difference of 1.89e-04 and their normal probability density function plot is almost identical as shown in Fig. 9(d)-(e); analysis of other methods, shown in Fig. 9(a)-(c), exhibit visible distortions in image histograms. This shows that the proposed method is more resistant to histogram attacks.

Table 2 Analysis of kurtosis and skewness for the respective cover and stego images obtained at 80% block modifications approximating to 1 bpp for all the algorithms

Image	Algorithm	Cover		stego	
		Kurtosis	Skewness	Kurtosis	Skewness
10.pgm	AE-LSBMR [24]	3.2978	1.1529	3.2953	1.1516
	Fuzzy Edge [20]			3.2972	1.1527
	Block Edge [2]			3.2970	1.1520
	BEASS method			3.2978	1.1529
1000.pgm	AE-LSBMR [24]	1.4039	0.0162	1.4029	0.0153
	Fuzzy Edge [20]			1.4007	0.0148
	Block Edge [2]			1.3999	0.0142
	BEASS method			1.4039	0.0162
1099.pgm	AE-LSBMR [24]	2.2717	0.3246	2.2722	0.3237
	Fuzzy Edge [20]			2.2713	0.3244
	Block Edge [2]			2.2676	0.3233
	BEASS method			2.2716	0.3245

The improved values against the respective parameters which shows the optimality of the method are shown in bold

4.2.3 RS steganalysis

RS analysis considers the placement of pixels in the stego image [10]. It is useful in detecting artifacts due to structural asymmetry introduced during the LSB embedding process. It does so by utilizing spatial correlations to design a discrimination function which divides the image into regular (R) and singular (S) groups and the total number of regular groups should be greater than singular groups. The zero-message hypothesis of RS steganalysis states that the values $RM \equiv R-M$ and $SM \equiv S-M$ i.e., RM should be approximately equal to R-M and SM should be approximately equal to S-M.

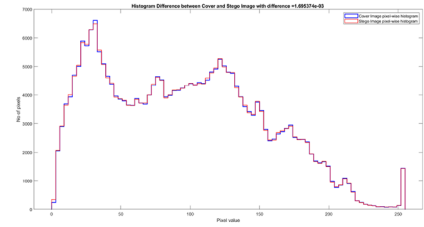
The RS steganalysis is unsuccessful in detecting the stego image as shown in Fig. 10. We begin with gradual embedding of message until embedding rate of 100% is reached, even then the difference between RM(SM) and R-M(S-M) values remains close and clears zero-message hypothesis.

4.2.4 Universal steganalysis

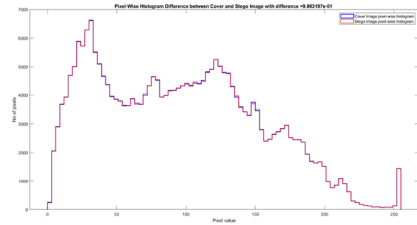
Most modern steganalyzers are challenged when modifications are made in image regions that necessitates designing complex statistical descriptors for the cover image. Universal/blind steganalysis are powerful and accurate tools designed to work with any steganographic algorithm, even with an unknown algorithm. These are built as supervised classifiers on feature vector, generally using support vector machine (SVM) known for its resistance against overtraining; that said, SVMs are highly complex to train, even for problems of moderate size.

To further test the robustness of the proposed method, we have used ensemble classifiers which are more flexible and scalable w.r.t. the number of training samples and feature dimensionality. The ensemble classifiers [18] consist of many base learners which are independently trained on subsamples of cover and stego images. Each of the base learner is

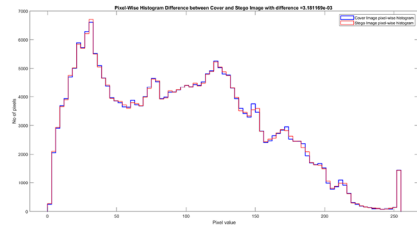
Fig. 9 Histogram Analysis of previous and proposed method of image 1099.pgm at 1 bpp embedding **a** AE-LSBMR, **b** Fuzzy Edge **c** Block Edge **d** BEASS method **e** BEASS method pdf of stego and cover



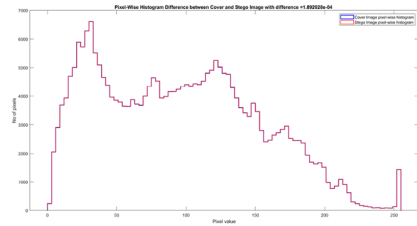
(a)



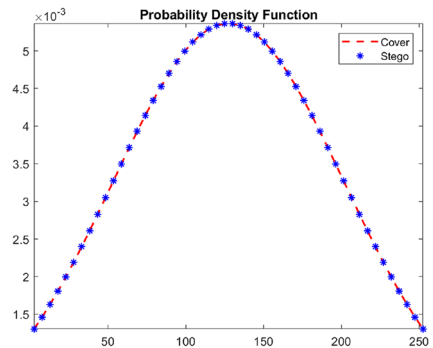
(b)



(c)



(d)



(e)

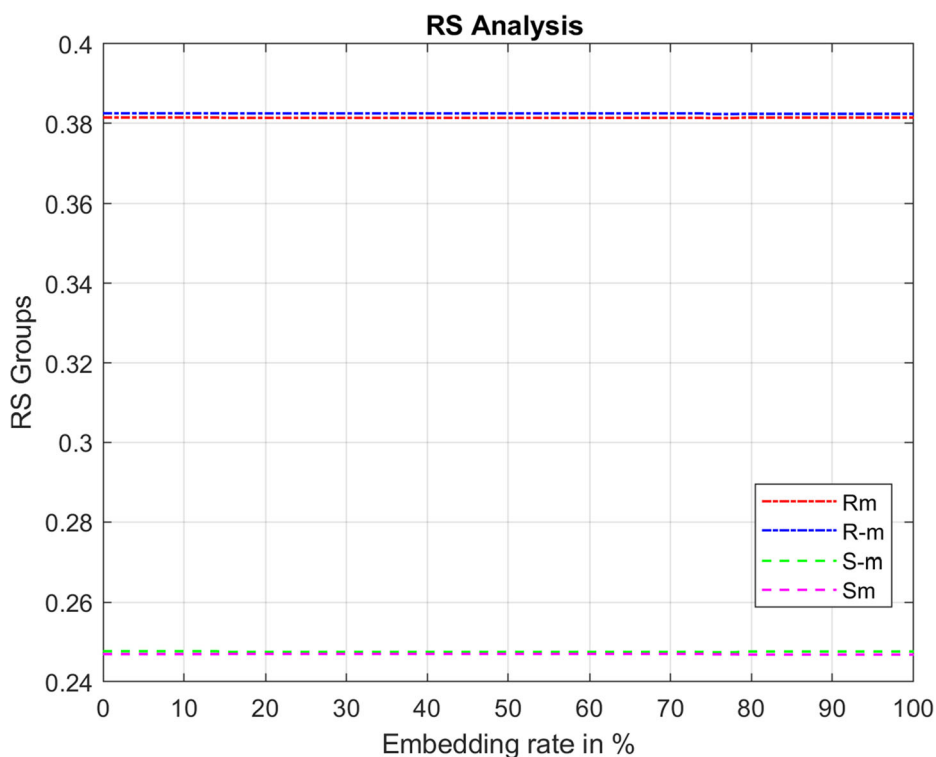


Fig. 10 RS diagram of 1099.pgm image post embedding. X-axis denotes the embedding rate and Y-axis denotes the relative change of regular and singular groups taken using Mask=[0 1 1 0]

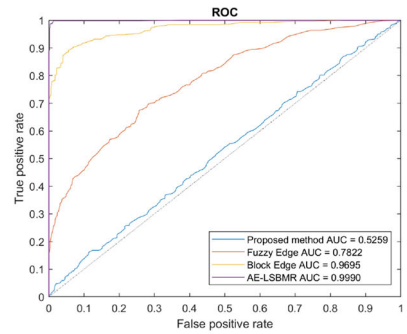
implemented using Fisher Linear Discriminant (FLD) and a final decision is made by aggregating the decisions of individual learners [15, 18].

Fig. 11(a-d) shows the experiment conducted on 1000 cover images, subjected to embedding block change of 10-80% respectively for all the four methods to generate 1000 stegos. To show that the proposed method cannot be detected using simple low-dimensional feature space, we generated SPAM second-order Markov model [26] of pixel differences exhibiting 686 dimensionalities for both the cover and stegos. The plotted ROC curve is used to test the steganalytic classifier or model in which TP rate is plotted along the Y-axis and FP rate is plotted along the X axis and the method with larger AUC area is considered most secure [8]. It is observed from Fig. 11 that BEASS has the largest AUC at all the embedding rates implying maximum tolerance against blind steganalysis attacks.

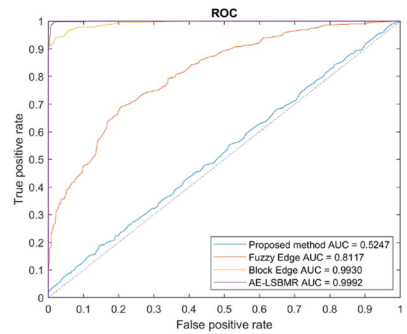
5 Conclusion

In this paper, a novel block-wise edge image steganography scheme (BEASS) in the spatial domain is presented. There are two major contributions of the proposed method. Firstly, the regions (blocks) to be embedded are adaptively determined using a measure of local complexity and the size of the payload, prioritizing regions with the maximum

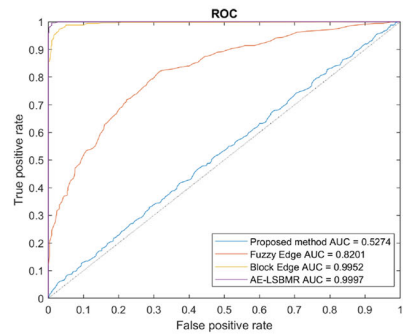
Fig. 11 ROC curves comparing the proposed method with three existing algorithms post embedding where x-axis represents the false positive rate and y-axis represents the true positive rate respectively, using ensemble classifier. **a** ROC curve at 0.3 bpp embedding rate, **b** ROC curve at 0.5 bpp embedding rate **c** ROC curve at 0.8 bpp embedding rate **d** 1 bpp embedding rate (80% block modifications)



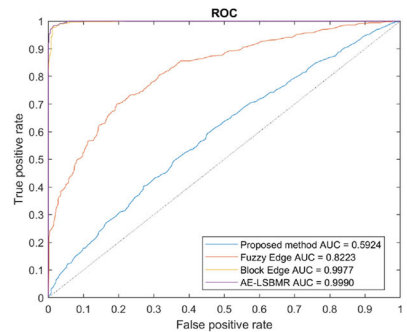
(a)



(b)



(c)



(d)

edge pixels. This reduces the block processing and embedding time since maximum amount of changes can be applied without introducing any visual distortion. Most edge steganography approaches facilitate high payloads by embedding in both edge and neighboring non-edge pixels but without analyzing the inter-pixel relationship, other methods result in severe statistical distortions that make detection easy and the algorithm insecure. The second contribution overcomes this limitation by embedding three message bits in the edge pixel and determines the safe embedding capacity for the neighboring non-edge pixel so that inter-pixel relationships and statistical resemblance is maintained. The experimental results from the evaluation of 1000 grayscale images from BOSSbase 1.01 at 80% embedding block change (1 bpp) show that the proposed method is visually imperceptible as determined from IQM with a PSNR of 61–65 db, SSIM in the range of 0.998 – 1, minimal KL-Divergence, Absolute Difference and NAE values. BEASS offers high payload embedding yet ensures statistical similarity as the kurtosis, skew of the cover and stego pdf remain undistorted. Furthermore, BEASS was found to be secure from histogram attack, RS steganalysis was deemed unsuccessful in discovering the hidden message and blind steganalysis attacks as the ROC curve of BEASS method indicated the largest AUC value in comparison with other methods at embedding rates of 0.3 to 1 bpp.

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